Choosing the Optimal Layover Buffer

Under Flight Delay Uncertainty

A Monte Carlo simulation approach to missed-connection risk

Primary message: 90 minutes is the best practical minimum buffer; 120 minutes is the conservative risk-minimizing choice.

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The Layover Trade-Off

Short layover

- Less waiting time
- More convenient itinerary
- Higher risk if the first flight is delayed

Decision tension

Long layover

- Lower missed-connection risk
- More reliable connection
- Higher time and opportunity cost

Decision under uncertainty

Business question
How much layover time should a traveler leave between flights?

What Decision Are We Making?

Choose layover buffer L
$L \in \{45, 60, 90, 120\}$

→

Arrival delay D is realized
first flight arrives late or on time

→

Connection outcome
$D > L \rightarrow$ missed
$D \leq L \rightarrow$ made

→

Scenario cost
Ticket + wait + missed penalty

Upfront decision

Made before the flight delay is known. The traveler commits to an itinerary / buffer.

Wait-and-see outcome

After D is observed, the traveler either makes or misses the connection.

Performance measure

Each buffer is evaluated by missed-connection probability, expected cost, and tail risk.

Flight Delays Are Zero-Inflated and Right-Skewed

Key uncertainty
D = arrival delay of the first flight

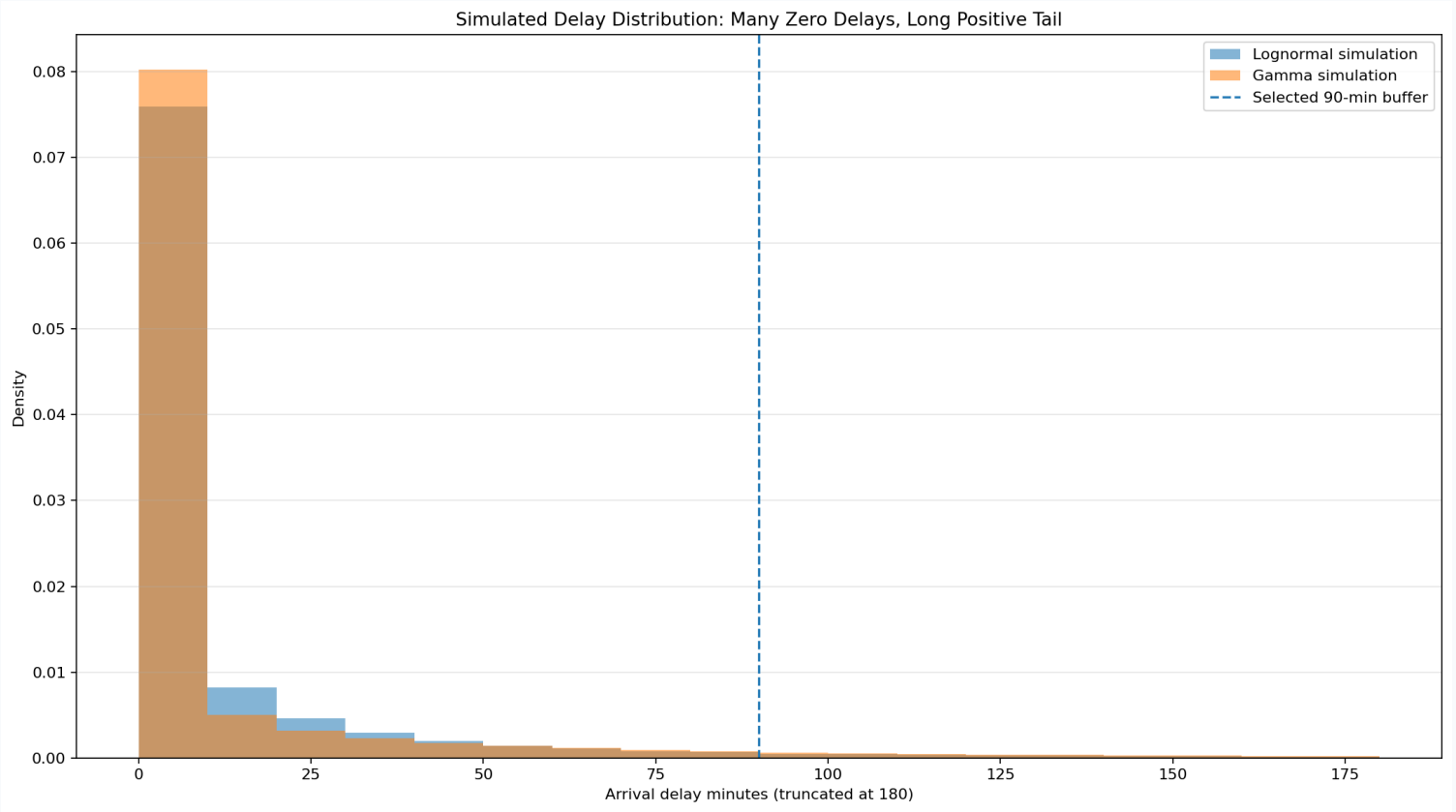
61.04%
No-delay probability

38.96%
Positive-delay probability

74 min
Historical P95 delay

176 min
Historical P99 delay

- Many observations have zero delay
- Positive delays have a long right tail
- Tail events drive missed-connection risk



Selected 90-min buffer line shown in chart

Two-Stage Decision Model

Uncertainty

D = nonnegative arrival delay
Zero-inflated lognormal / gamma

Decision

$L \in \{45, 60, 90, 120\}$
chosen before D is known

Wait-and-see

$M_L(D) = 1$ if $D > L$
otherwise 0

Objective

Minimize $E[\text{Total Cost}_L] + \lambda \times \text{CVaR95}(\text{Total Cost}_L)$
$\text{Total Cost}_L = 350 + (L / 60) \times 40 + 650 \times M_L(D)$

Constraints

- $P(D > L) \leq 5\%$
- Ticket cost ≤ budget
- L must be one of four buffers
- Scenario logic: miss only if $D > L$

Base parameters

Input	Value
Ticket cost	$350
Value of time	$40/hr
Missed penalty	$650
λ	0.5
α	0.05

SOLUTION METHOD

Monte Carlo Simulation Approach

1

Fit delay distributions

Estimate zero-inflated lognormal and gamma models from cleaned historical delay data.

2

Simulate outcomes

Generate 10,000 possible first-flight arrival delays for each distribution.

3

Evaluate buffers

Test 45, 60, 90, and 120 minutes against every simulated delay.

4

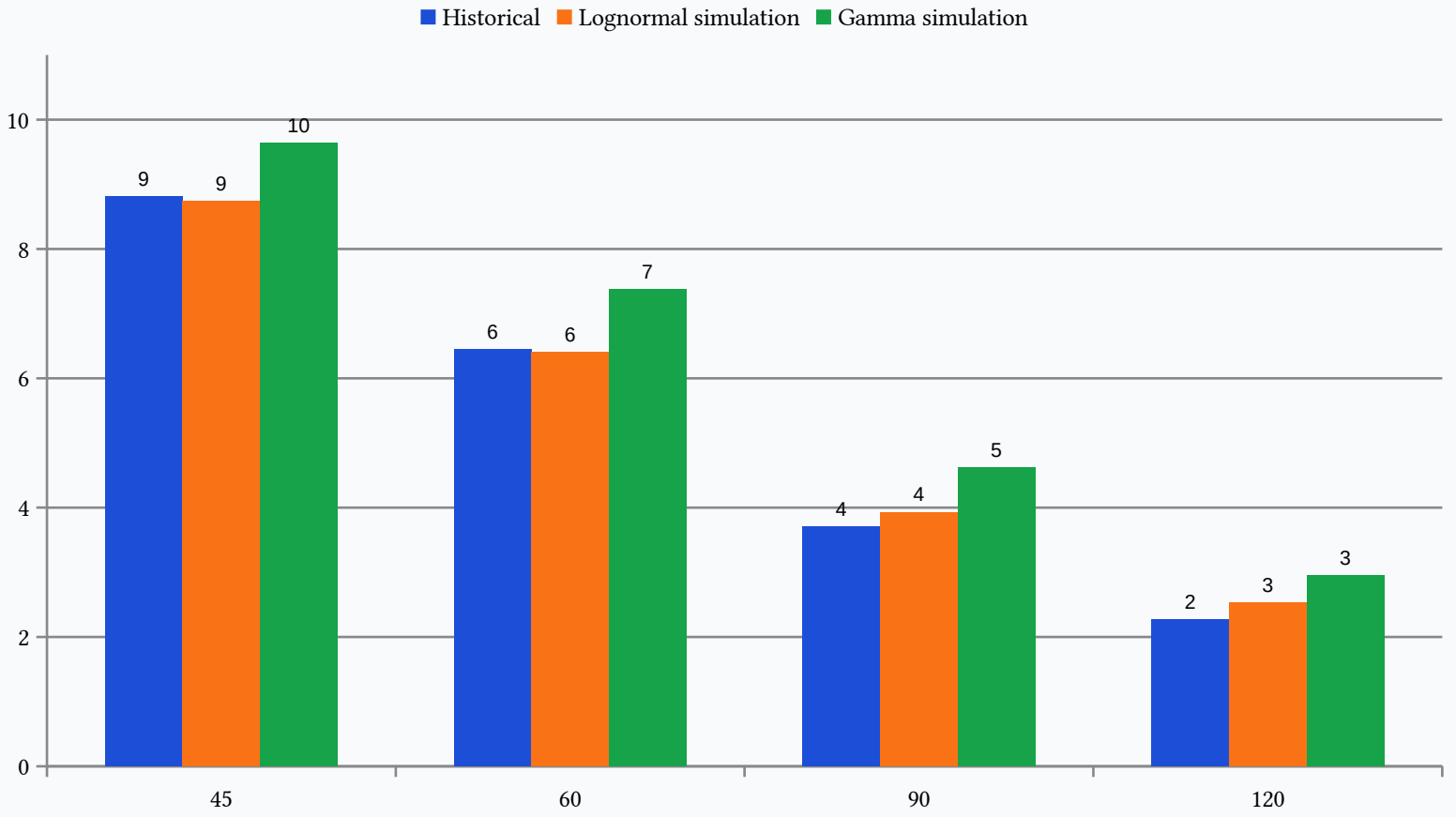
Compare decisions

Calculate risk, expected cost, VaR/CVaR, and risk-adjusted cost.

Simulation output feeds the recommendation: choose a buffer that balances waiting cost and missed-connection risk.

KEY RESULT 1

Longer Buffers Reduce Missed-Connection Risk



5% risk tolerance

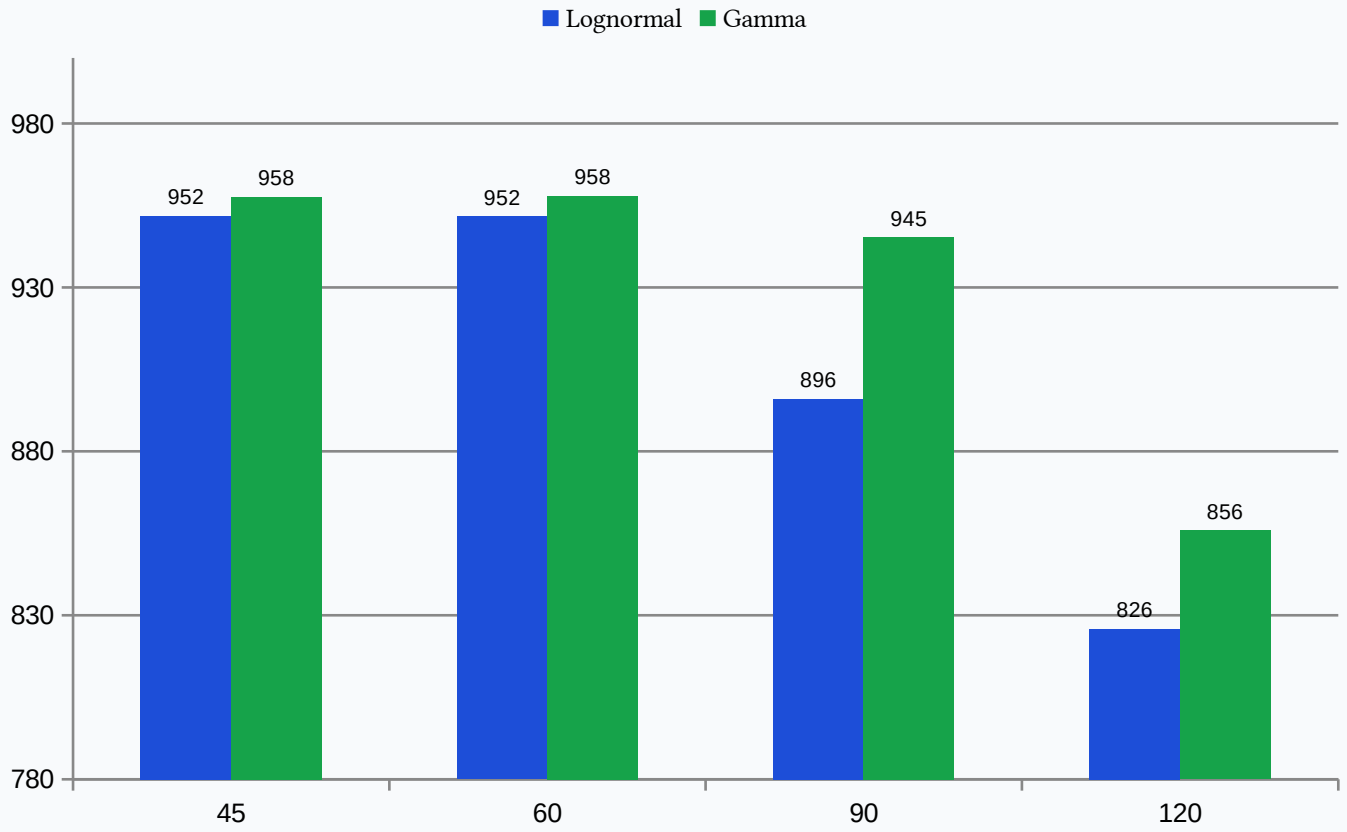
Main insight

- 45 and 60 minutes exceed the 5% tolerance.
- 90 minutes is the first buffer below 5% across both fitted distributions.
- 120 minutes is safer, but adds more waiting time.

Therefore, 90 minutes is the minimum service-level feasible buffer.

KEY RESULT 2

Expected Cost vs. Tail Risk



Lognormal model base-case results

Buffer	Exp. Cost	CVaR95	Risk Adj.
45	$436.81	$1,030	$951.81
60	$431.67	$1,040	$951.67
90	$435.55	$920.90	$896.00
120	$446.45	$758.90	$825.90

Interpretation

- 60 min has the lowest expected cost, but violates the 5% risk tolerance.
- 120 min has the lowest tail risk and risk-adjusted cost.
- 90 min is the best practical minimum buffer when we balance risk and waiting time.

RECOMMENDATION

90 Minutes for Typical Travelers; 120 Minutes for High-Stakes Trips

Primary recommendation

90-minute layover

- First option that satisfies $P(D > L) \leq 5\%$
- Avoids the high missed-connection risk of 45/60 minutes
- Avoids the extra waiting time of 120 minutes

Typical traveler

Conservative recommendation

120-minute layover

- Lowest missed-connection probability
- Lowest CVaR and risk-adjusted cost
- Best for hard deadlines or highly risk-averse travelers

Business traveler / hard deadline

Final message: choose 90 minutes as the practical minimum; upgrade to 120 minutes when the cost of disruption is unusually high.

What We Simplified and How the Model Could Improve

Current model limitations

- Ticket cost fixed at $350 in the base case
- Missed-connection cost fixed at $650
- Does not model cancellations, diversions, gate distance, or terminal transfer time
- Uses an aggregate delay distribution rather than route-specific estimates

Future work

- Add real itinerary-specific ticket prices
- Model missed-connection cost as a distribution
- Condition delay by airline, airport, month, day of week, and departure hour
- Add actual minimum connection times and terminal transfer times

Takeaway

Even with a simplified model, simulation gives a clear and actionable decision rule: use data to avoid underestimating tail-risk in connecting flights.